ALGEBRA

बीजगणित

SHEET -03

CLASS ROOM SHEETS / FOR ALL EXAMS

UPDATED

ALGEBRA (बीजगणित)

CLASSROOM SHEET-03

CONCEPT OF SYMMETRY

1. If $xy + yz + zx = 1$, then $\frac{1+y^2}{(x+y)(y+z)}$ = ?

(a) 0 (b) 1
(c) 2 (d) 3

2. If $x^2 + y^2 + z^2 = xy + yz + zx$, then the value of $\left(\frac{17x^4+9y^4+16z^4}{8x^2y^2+6y^2z^2+10z^2x^2}\right)$ is:

SSC Phase-XII 24/06/2024 (Shift-02)

(a) 1.75 (b) 3.25
(c) 2.00 (d) 2.50

3. If $a^2 + b^2 + c^2 = ab + bc + ca$ then $\frac{a+c}{b}$ is

(a) 0 (b) 2
(c) 1 (d) –1

4. If $\frac{4x-3}{x} + \frac{4y-3}{y} + \frac{4z-3}{z} = 0$ then $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ is

(a) 9 (b) 3
(c) 4 (d) 6

5. If $\frac{2+a}{a} + \frac{2+b}{b} + \frac{2+c}{c} = 4$ then the value of $\frac{(ab+bc+ca)}{abc}$ is

(a) 2 (b) 1
(c) 0 (d) $\frac{1}{2}$

6. If $\frac{x+a^2+2c^2}{b+c} + \frac{x+b^2+2a^2}{c+a} + \frac{x+c^2+2b^2}{a+b} = 0$, find x

(a) $a^2 + b^2 + c^2$ (b) $-(a^2 + b^2 + c^2)$
(c) $a^2 + 2b^2 + c^2$ (d) $-(a^2 + 2b^2 + c^2)$

7. If $\frac{x-a^2}{b^2+c^2} + \frac{x-b^2}{c^2+a^2} + \frac{x-c^2}{b^2+a^2} = 3$, find the value of x.

(a) $a^2 + b^2 - c^2$ (b) $a^2 + b^2 + c^2$
(c) $a^2 - b^2 - c^2$ (d) $a^2 + b^2$

8. If $bc + ca + ab = abc$ then

$\frac{b+c}{bc(a-1)} + \frac{c+a}{ca(b-1)} + \frac{a+b}{ab(c-1)}$ = ?

(a) 0 (b) 1
(c) 2 (d) 3

9. If $a^2 = b + c$, $b^2 = c + a$, $c^2 = a + b$, then

$\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c}$

(a) 0 (b) 1
(c) 2 (d) 3

10. If a, b, c are non zero, $a+\frac{1}{b} = 1$ & $b+\frac{1}{c} = 1$ then (i) abc is (ii) $c + \frac{1}{a}$ is

(a) –1, 1 (b) 3, –1
(c) –3, 1 (d) 1, 1

11. If $a^x = (x + y + z)^y$, $a^y = (x + y + z)^z$ and $a^z = (x + y + z)^x$, then $x + y + z$ = ? $(a \neq 0)$

(a) 0 (b) 1
(c) a^3 (d) a

12. If $a + b + c = 0$, then the value of $\frac{a^2+b^2+c^2}{a^2-bc}$

(a) 0 (b) 1
(c) 2 (d) 3

13. If $a + b + c = 0$ then the value of $\frac{a^2-bc}{b^2-ca}$ is :

(a) 0 (b) 1
(c) 2 (d) 3

14. If $a + b + c = 0$ then the value of $\frac{a^2+b^2+c^2}{ab+bc+ca}$ is:

(a) 2 (b) –2
(c) 0 (d) 4

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CONCEPT OF VALUE PUTTING

15. If $x + y = 1$, then what is the value of $x^3 + 3xy + y^3$?

SSC CGL MAINS (08/08/2022)

(a) 0 (b) 2
(c) 1 (d) –1

16. If $a + b = 6$ and $ab = 5$, then what is the value of $a^3 + b^3$?

SSC CGL MAINS (08/08/2022)

(a) 136 (b) 126
(c) 116 (d) 106

17. If $a^3 + b^3 = 217$ and $a + b = 7$, then the value of ab is:

SSC CPO 24/11/2020 (Shift- 02)

(a) –6 (b) –1
(c) 7 (d) 6

18. If $r + \frac{64}{r} = 16$, then the value of $r^4 + \frac{1}{r^3}$ is.

SSC CHSL 25/05/2022 (Shift- 03)

(a) $4096\frac{1}{512}$ (b) 512
(c) $512\frac{1}{4096}$ (d) 4096

19. If $a - \frac{12}{a} = 1$, $a > 0$, find $a^2 + \frac{16}{a^2}$?

(a) 15 (b) 19
(c) 17 (d) 11

20. If $(x+6y) = 8$, and $xy = 2$, where $x > 0$, what is the value of $(x^3 + 216\,y^3)$?

SSC CGL 11/04/2022 (Shift- 01)

(a) 288 (b) 224
(c) 368 (d) 476

21. If $a^2 + b^2 = 65$ and $ab = 8$, $a > b > 0$, then find the value of $a^2 - b^2$.

SSC CGL 21/04/2022 (Shift- 02)

(a) 65 (b) 60
(c) 72 (d) 63

22. If $x^2 + 9y^2 = 40$ and $xy = 4$, where $x > 0$, $y > 0$, then what is the value of $(x^3 + 27y^3)$?

SSC PHASE IX 2022

(a) 224 (b) 416
(c) 440 (d) 800

23. If $xy = -6$ and $x^3 + y^3 = 19$ (x and y are integers), then what is the value of $\frac{1}{x^{-1}} + \frac{1}{y^{-1}}$?

SSC CGL MAINS (08/08/2022)

(a) –2 (b) 2
(c) –1 (d) 1

24. If $2a + 2b + c = 0$, then find the value of $\frac{4a^2 + 4b^2 + 4c^2}{5c^2 - 8ab}$.

SSC CPO 29/06/2024 (Shift-03)

(a) –1 (b) $-2\sqrt{3}$
(c) 1 (d) $2\sqrt{3}$

25. If $x - y + z = 0$, then find the value of $\frac{y^2}{2xz} - \frac{x^2}{2yz} - \frac{z^2}{2xy}$.

SSC CGL 12/04/2022 (Shift- 03)

(a) $\frac{3}{2}$ (b) $\frac{1}{2}$
(c) –6 (d) $-\frac{3}{2}$

26. If $a + b + c = 0$, then what is the value of $\frac{(b+c)^2}{bc} + \frac{(c+a)^2}{ca} + \frac{(a+b)^2}{ab}$?

SSC CGL 18/08/2021 (Shift- 03)

(a) 1 (b) –3
(c) –1 (d) 3

27. If $a + b - c = 0$, then what is the value of $\frac{(b-c)^2}{4bc}\frac{(c-a)^2}{4ca}\frac{(a+b)^2}{4ab}$?

SSC CHSL 06/08/2021 (Shift- 01)

(a) $-\frac{3}{4}$ (b) $\frac{3}{4}$
(c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

28. If $x + y + z = 0$, then what will be the value of $\left(\frac{x^2}{yz}\right) + \left(\frac{y^2}{zx}\right) + \left(\frac{z^2}{xy}\right)$?

SSC CHSL 24/05/2022 (Shift- 01)

(a) $\frac{3(x^2+y^2+z^2)}{xyz}$ (b) $x^2 + y^2 + z^2$
(c) $\frac{x^2y^2z^2}{x}$ (d) 3

29. If $a + b + c = 0$, then $\left(\frac{a+b}{c} + \frac{b+c}{a} + \frac{c+a}{b}\right)\left(\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}\right)$ is :

(a) 8 (b) 9
(c) – 3 (d) 0

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30. If $x = \frac{a}{b} + \frac{b}{a}$, $y = \frac{b}{c} + \frac{c}{b}$, $z = \frac{c}{a} + \frac{a}{c}$, then what is the value of $xyz - x^2 - y^2 - z^2$ = ?

(a) –4 (b) –2
(c) –1 (d) –6

31. If $a^3b = abc = 180$, a, b, c are positive integers, then the value of c is :

(a) 110 (b) 1
(c) 4 (d) 25

32. If $x = a + \frac{1}{a}$ and $y = a - \frac{1}{a}$ then $\sqrt{x^4 + y^4 - 2x^2y^2}$ is equal to :

(a) $16a^2$ (b) 8
(c) $\frac{8}{a^2}$ (d) 4

33. If $x = 2 - p$, then $x^3 + + 6xp + p^3$ is equal to :

(a) 12 (b) 6
(c) 8 (d) 4

34. Find the product of $(a + b + 2c)(a^2 + b^2 + 4c^2 - ab - 2bc - 2ca)$.

SSC CGL 07/03/2020 (Shift- 03)

(a) $a^3 + b^3 + 8c^3 - 6abc$
(b) $a^3 + b^3 + 6c^3 - 6abc$
(c) $a^3 + b^3 + 8c^3 - 2abc$
(d) $a^3 + b^3 + 8c^3 - abc$

35. If a + b – c = 5 and ab – bc – ac = 10, then find the value of $a^2 + b^2 + c^2$.

SSC CGL 18/04/2022 (Shift- 01)

(a) 15 (b) 45
(c) 5 (d) 40

36. If a + b + c = 11 and ab + bc + ca = 28, then find the value of $a^3 + b^3 + c^3 - 3abc$.

SSC CGL 19/04/2022 (Shift- 03)

(a) 407 (b) 1639
(c) 2255 (d) 1093

Miscellaneous Questions

37. If p = 11, then the value of $p(p^2 + 3p + 3)$ is:

SSC CHSL 25/05/2022 (Shift- 03)

(a) 1629 (b) 1225
(c) 1727 (d) 1111

38. If p = 38, then the value of $p(p^2 + 3p + 3)$ is _____.

SSC CHSL 06/06/2022 (Shift- 03)

(a) 39313 (b) 59319
(c) 39318 (d) 59318

39. If $x = 11$, then the value of $(x^5 - 12x^4 + 12x^3 - 12x^2 + 12x - 1)$ is:

SSC CPO 28/06/2024 (Shift-03)

(a) 10 (b) 8
(c) 9 (d) 14

40. If $a^4 + b^4 + a^2b^2 = 273$ and $a^2 + b^2 - ab = 21$, then one of the values of $\left(\frac{1}{a} + \frac{1}{b}\right)$ is:

SSC CGL 24/08/2021 (Shift- 01)

(a) $-\frac{9}{4}$ (b) $-\frac{3}{4}$
(c) $\frac{9}{8}$ (d) $\frac{3}{2}$

41. If $x^4 + y^4 + x^2y^2 = 17\frac{1}{16}$ and $x^2 - xy + y^2 = 5\frac{1}{4}$, then one of the values of (x – y) is:

SSC CGL MAINS 03/02/2022

(a) $\frac{5}{2}$ (b) $\frac{3}{4}$
(c) $\frac{5}{4}$ (d) $\frac{3}{2}$

42. $ab(a - b) + bc(b - c) + ca(c - a)$ is equal to :

(a) $(a + b)(b - c)(c - a)$
(b) $(a - b)(b + c)(c - a)$
(c) $(a - b)(b - c)(c - a)$
(d) $(b - a)(b - c)(c - a)$

43. If $x^4 + x^2y^2 + y^4 = 273$ and $x^2 - xy + y^2 = 13$, then the value of xy is :

SSC CGL 05/03/2020 (Shift- 02)

(a) 4 (b) 8
(c) 10 (d) 6

44. If $16a^4 + 36a^2b^2 + 81b^4 = 91$ and $4a^2 + 9b^2 - 6ab = 13$, then what is the value of $3ab$?

SSC CGL 04/03/2020 (Shift- 01)

(a) – 3 (b) $\frac{3}{2}$
(c) 5 (d) $-\frac{3}{2}$

45. If $\frac{7(x^2 + 1) - 17x}{7x} = 2$, then $\left(\frac{x + 1}{\sqrt{x}}\right)^2$ = ?

(a) 6.45 (b) 6.53
(c) 6.86 (d) 6.31

46. If $\frac{7\left(2\sqrt{2} - x\right)}{x} = 7 + \frac{14}{x^2}$ then $\sqrt{\left(23x^{64} + 42x^{12} + 163\right)}$

(a) 12 (b) $12\sqrt{2}$
(c) $12\sqrt{3}$ (d) 24

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47. If $8k^6 + 15k^3 - 2 = 0$, then the positive value of $\left(k + \frac{1}{k}\right)$ is:

SSC CGL 12/04/2022 (Shift- 03)

(a) $2\frac{1}{2}$ (b) $2\frac{1}{8}$

(c) $8\frac{1}{2}$ (d) $8\frac{1}{8}$

48. If $(ab + bc + ca) = 0$, then what is the value of $\left(\frac{1}{a^2 - bc} + \frac{1}{b^2 - ca} + \frac{1}{c^2 - ab}\right)$?

SSC CHSL 09/06/2022 (Shift- 01)

(a) 2 (b) 0

(c) 1 (d) $a + b + c$

49. If $px^3 + x^2 + 3x + q$ is exactly divisible by $(x + 2)$ and $(x - 2)$, then the values of p and q are:

SSC CHSL 08/06/2022 (Shift- 03)

(a) $p = -\frac{3}{4}$ and $q = 4$

(b) $p = \frac{3}{4}$ and $q = 4$

(c) $p = \frac{3}{4}$ and $q = -4$

(d) $p = -\frac{3}{4}$ and $q = -4$

50. What is the value of the following expression?

$$2^2\left(\frac{x^a}{x^b}\right)^{(a+b)} \times 3^2\left(\frac{x^b}{x^c}\right)^{(b+c)} \times 6^{-2}\left(\frac{x^c}{x^a}\right)^{(a+c)}$$

SSC CHSL 08/06/2022 (Shift- 02)

(a) 1 (b) 0

(c) 4 (d) 9

51. Simplify the following expression.

$$\frac{5(a^6 - b^6)^3 + 5(b^6 - c^6)^3 + 5(c^6 - a^6)^3}{2(a^3 - b^3)^3 + 2(b^3 - c^3)^3 + 2(c^3 - a^3)^3}$$

SSC CGL 24/08/2021 (Shift- 02)

(a) $\frac{5}{2}(a^3 + b^3)(b^3 + c^3)(c^3 + a^3)$

(b) $\frac{5}{2}(a^3 + b^3)(b^3 - c^3)(c^3 - a^3)$

(c) $\frac{5}{2}(a^3 - b^3)(b^3 + c^3)(c^3 + a^3)$

(d) $\frac{5}{2}(a^3 - b^3)(b^3 - c^3)(c^3 + a^3)$

52. If $x^3 + 4x = 8$, find $x^7 + 64x^2$?

(a) 32 (b) 64

(c) 128 (d) 256

53. If $x^2 - 5x + 25 = 0$ then $x^5 - 5x^4 - 3100$?

(a) 25 (b) $25\sqrt{2}$

(c) $25\sqrt{3}$ (d) None

54. If $(x + 5)(x + 6) + \frac{9}{x(x - 1)} = 0$, Find $3x^2 + 15x - 1$?

(a) 4 (b) 6

(c) 8 (d) 10

55. If $ax + by = 8$, $ay - bx = 6$, $a^2 + b^2 + x^2 + y^2 = 29$

Find $\frac{a^2 + b^2}{(x^2 + y^2)}$; $a^2 + b^2 > x^2 + y^2$?

(a) $\frac{25}{4}$

(b) $\frac{27}{4}$

(c) $\frac{23}{4}$

(d) $\frac{29}{4}$

56. If $\frac{a^2 + b^2 + c^2 - 1024}{ab - bc - ca} = -2$ and $a + b = 5c$, where $c > 0$, then the value of c is _____.

(a) 8 (b) 4

(c) 12 (d) 5

57. If $a^3 - b^3 = 602$ and $a - b = 2$, then find the value of $(a^2 + b^2)$.

IB ACIO GRADE II 17/01/2024 (Shift-01)

(a) 156

(b) 240

(c) 202

(d) 260

58. If $x = 7 + 7^{\frac{2}{3}} + 7^{\frac{1}{3}}$, then which of the options below is correct.

IB ACIO GRADE II 17/01/2024 (Shift-02)

(a) $x^3 + 21x^2 - 126x - 252 = 0$

(b) $x^3 + 21x^2 - 126x + 252 = 0$

(c) $x^3 - 21x^2 - 126x + 252 = 0$

(d) $x^3 - 21x^2 + 126x - 252 = 0$

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59. **Select the option that is true regarding the following labelled Assertion (A) and Reason (R).**

Assertion (A): The value of

$$\frac{\left[(3.9)^3+9\times1.3\times4.29+11.7\times(1.1)^2+1.331\right]}{\left[1.23\times1.23+0.77(0.77+0.6\times4.1)\right]}$$

is 31.25.

Reason (R): Using $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$ and $(x+y)^2 = x^2 + y^2 + 2xy$

IB ACIO GRADE II 18/01/2024 (Shift-01)

(a) A is true and R is false.
(b) Both A and R are true but R is not a correct explanation of A.
(c) Both A and R are true and R is a correct explanation of A.
(d) A is false and R is true.

60. **If x, y and z are positive numbers and $x + y + z = 1$, then the least value of $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}$ is:**

SSC CGL 09/09/2024 (Shift-03)

(a) 7 **(b) 5**
(c) 9 **(d) 11**

ANSWER KEY

1.(b)	2.(a)	3.(b)	4.(c)	5.(d)	6.(b)	7.(b)	8.(b)	9.(b)	10.(a)
11.(d)	12.(c)	13.(b)	14.(b)	15.(c)	16.(b)	17.(d)	18.(a)	19.(c)	20.(b)
21.(d)	22.(a)	23.(d)	24.(c)	25.(a)	26.(d)	27.(a)	28.(d)	29.(b)	30.(a)
31.(b)	32.(d)	33.(c)	34.(a)	35.(c)	36.(a)	37.(c)	38.(d)	39.(a)	40.(b)
41.(a)	42.(d)	43.(a)	44.(d)	45.(a)	46.(a)	47.(a)	48.(b)	49.(d)	50.(a)
51.(a)	52.(c)	53.(a)	54.(c)	55.(a)	56.(a)	57.(c)	58.(d)	59.(c)	60.(c)

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CALCULATION TECHNIQUES

TECHNIQUE – 01



♦ Square Root of perfect square ♦

Calculate the square root of 5184

Step 01: The unit digit of 5184 is 4, so its square root will be end on 2 or 8.

Step 02: Now leave two digits from end and check the remaining number. In this, case it is 51.

$7^2 < 51 < 8^2$

So, the tens digit of square root is 7.

Step 03: Now the square root of 5184 can be 72 or 78. For this, we have to check whether 51 is closer to 7^2 or 8^2 As 51 is closer to 7^2.

Square root of 5184 is 72.



♦ Do it Yourself ♦

(i) $\sqrt{7744}$ = ...

(ii) $\sqrt{1764}$ = ...

(iii) $\sqrt{6561}$ = ...

(iv) $\sqrt{7396}$ = ...

(v) $\sqrt{9216}$ = ...

(vi) $\sqrt{4624}$ = ...

(vii) $\sqrt{7569}$ = ...

TECHNIQUE – 02



♦ Square Root of Non-perfect square ♦

$$\sqrt{N} = \sqrt{(A \pm B)} = \sqrt{A} \pm \frac{B}{2\sqrt{A}}$$

(First find a number which is perfect square near to the number N)

Ex: $\sqrt{10} = \sqrt{9+1} = \sqrt{9} + \frac{1}{2\sqrt{9}} = 3 + \frac{1}{2\times 3} = \frac{19}{6} = 3.17$

Ex: $\sqrt{14} = \sqrt{16-2} = \sqrt{16} - \frac{2}{2\sqrt{16}} = 4 - \frac{2}{2\times 4} = \frac{30}{8} = 3.75$



♦ Do it Yourself ♦

(i) $\sqrt{18}$ = ...

(ii) $\sqrt{29}$ = ...

(iii) $\sqrt{57}$ = ...

(iv) $\sqrt{265}$ = ...

(v) $\sqrt{567}$ = ...

SELF MENTORSHIP TABLE

	STEP-01	STEP-02	STEP-03	STEP-04	STEP-05	STEP-06
TOPIC NAME	VIDEOS	CLASSROOM SHEET	SSC MATHS	NOTES / FORMULA BOOK REVISION	CHAPTER TEST	PRO LEVEL PRACTICE SHEET
ALGEBRA SHEET-03						
CONCEPT OF SYMMETRY						
CONCEPT OF VALUE PUTTING						
Miscellaneous Questions						